THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A leaf from a plant kept in the dark and another from sunlight are tested with iodine. The one that turns blue-black is: 1

(a) the sunlight leaf **(b)** the dark leaf
(c) both **(d)** neither

2. A ray of light strikes a plane mirror at an angle of 40° to the normal. The angle between the incident and reflected rays is: 1

(a) 40° **(b)** 80° **(c)** 50° **(d)** 100°

3. If the Earth stopped rotating, the part facing the Sun would: 1

(a) have rain **(b)** have continuous daylight
(c) cool down **(d)** have night

4. A potted plant is covered with a polythene bag and kept in sunlight. After some time water droplets are seen inside the bag. This shows: 1

(a) respiration **(b)** transpiration **(c)** digestion **(d)** photosynthesis

5. A tree casts a shadow of 10 m when a 2 m stick casts a 1 m shadow at the same time. The height of the tree is: 1

(a) 5 m **(b)** 12 m **(c)** 10 m **(d)** 20 m

6. The Moon's phases when seen from the northern hemisphere: waxing crescent appears in the: 1

(a) morning sky only **(b)** evening sky (west)
(c) midnight **(d)** noon

7. Cuscuta has no chlorophyll. It survives by: 1

(a) taking food from its host **(b)** absorbing from dead matter
(c) eating insects **(d)** making food

8. In a pinhole camera, if the object distance is doubled, the image size becomes: 1

(a) same **(b)** quarter **(c)** half **(d)** double

9. During a total solar eclipse, the observer is in the Moon's: 1

(a) umbra **(b)** shadow of Earth **(c)** sunlight **(d)** penumbra

10. In the variegated leaf, the white parts are tested with iodine. They: 1

(a) remain brown **(b)** turn green **(c)** turn blue-black **(d)** turn red

11. A ray strikes the mirror surface making an angle of 25° with the surface. The angle of reflection is: 1

(a) 25° **(b)** 90° **(c)** 65° **(d)** 50°

12. If the Earth's axis were not tilted, we would have: 1

(a) no seasons **(b)** no day **(c)** more seasons **(d)** no night

13. Which plant cell part controls the opening of stomata? 1

(a) phloem **(b)** root hair **(c)** xylem **(d)** guard cells

14. The shadow of a pole is shortest at: 1

(a) night **(b)** morning **(c)** noon **(d)** evening

15. A full moon rises at about: 1

(a) sunrise **(b)** sunset **(c)** midnight **(d)** noon

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are potted plants kept near a window grown healthier than those kept in the dark? 2

17. The angle of incidence of a ray on a plane mirror is 35°. Find the angle of reflection and the angle between incident and reflected rays. 2

18. Why do days and nights have unequal lengths in different seasons? 2

19. Why do plants have many stomata on the lower surface of the leaf? 2

20. Why do we use a white screen for a pinhole camera image? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Design an experiment to show that CO2 is needed for photosynthesis. 3

22. A ray of light makes an angle of 70° with the surface of a plane mirror. Find the angle of incidence, reflection and the angle between incident and reflected rays. 3

23. Explain with a diagram in words how a total solar eclipse is formed. 3

24. Explain how stomata open and close. 3

OR

Why do we use a mirror at the corner of a blind road? 3

25. Describe the phenomenon of midnight Sun. 3

OR

Describe how a student can show that oxygen is released in photosynthesis. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain, with a labelled description, how a leaf is adapted for photosynthesis. 5

OR

A ray of light strikes a plane mirror at 60° to the surface. Draw and describe the path of the reflected ray and find all angles. 5

27. Draw (describe) the positions of the Earth in its orbit at the two solstices and the two equinoxes and write the seasons in both hemispheres. 5

OR

Describe the experiment on transpiration with a polythene bag and the precautions. 5

END OF QUESTION PAPER **Total: 50 Marks**

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